

Technical change and health: Evidence from the reduction of sugarcane burning in Brazil

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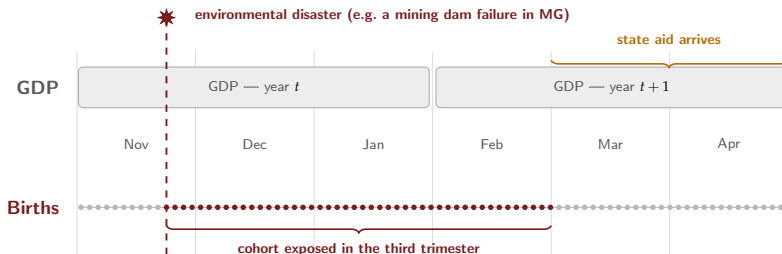
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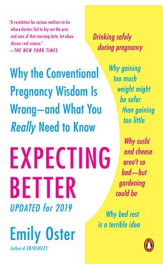
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Health at birth: a thermometer that reads every day



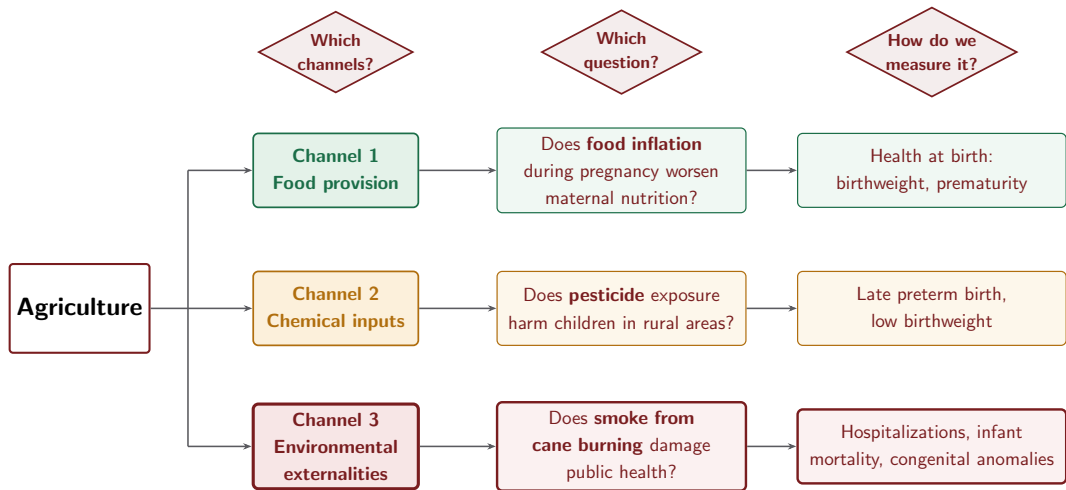
- GDP is a great measure of development, but it arrives once a year
- Babies are born all the time — in 2024, **64** a day in Salvador, **46** in Curitiba, **2** even in Viçosa-MG



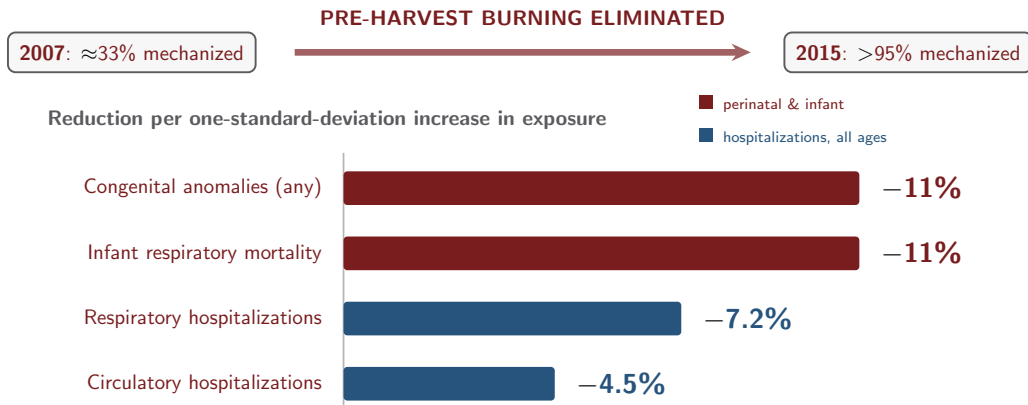
Emily Oster (Brown),
Expecting Better (2013)

Birth records are high-frequency, precisely dated, and cover every birth — a thermometer of local conditions.

How does agriculture affect health?



Can a technical change in agriculture improve health?

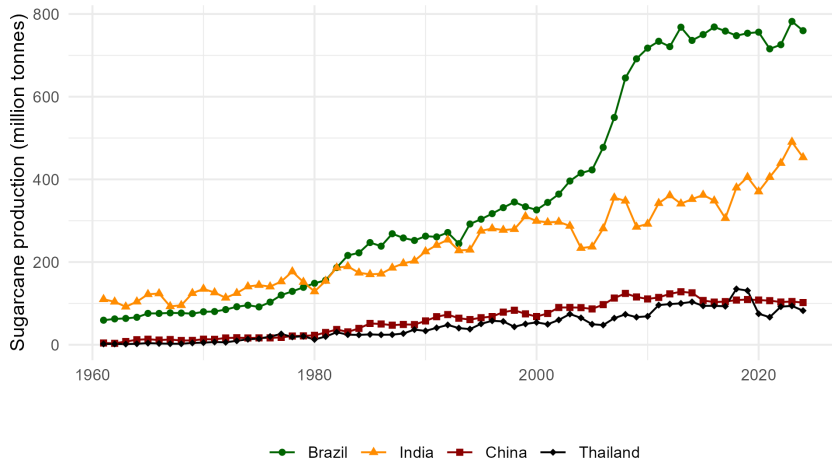


And it really is the smoke:

✓ pollution fell **only** in harvest months

✓ **downwind** municipalities gained most

Context

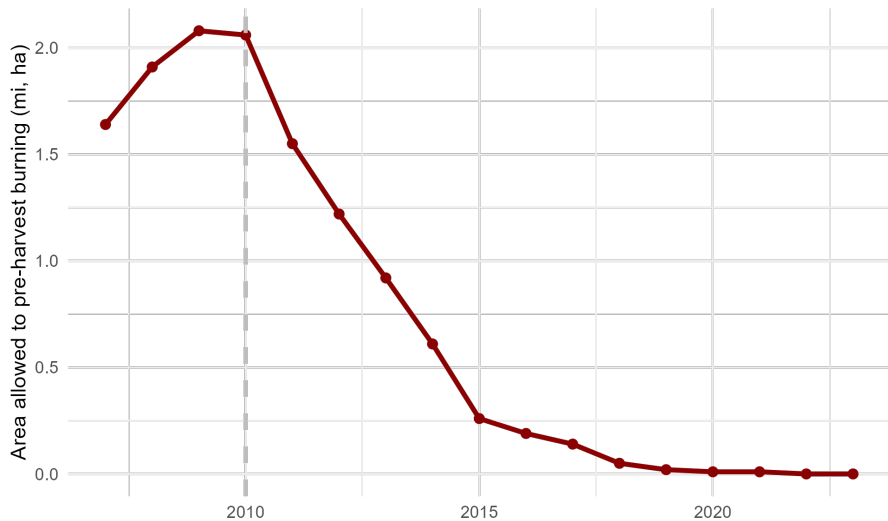


Our Source of Identification

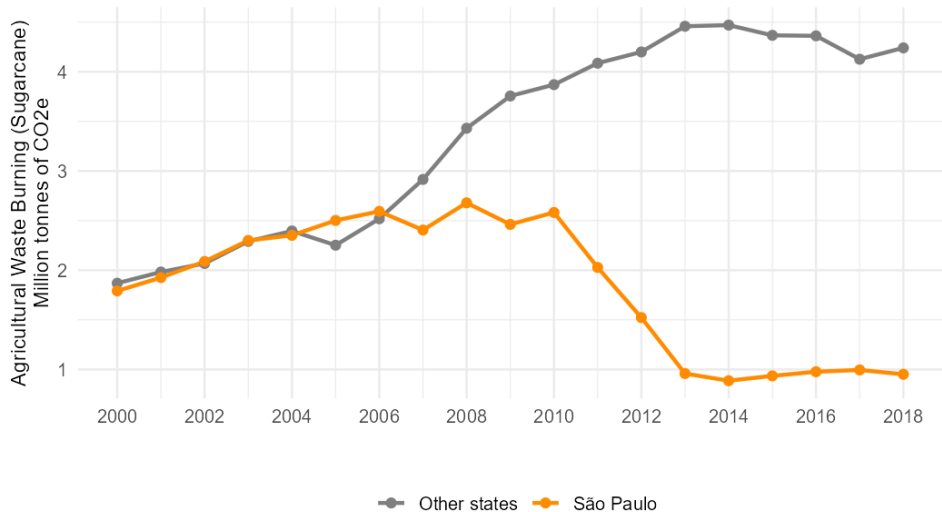
There was a reduction in burning driven by...

- Law
 - ▶ State Law 11.241/2002: gradual ban on straw burning (targets: 50% by 2011, 100% by 2021)
- Protocol
 - ▶ Green Ethanol Protocol (2007): voluntary industry agreement that accelerated deadlines to 2014–2017 in exchange for preferential financing and export market access
- Labor costs
 - ▶ Rising rural wages and export sustainability standards reinforced the shift [Davis, 2017]

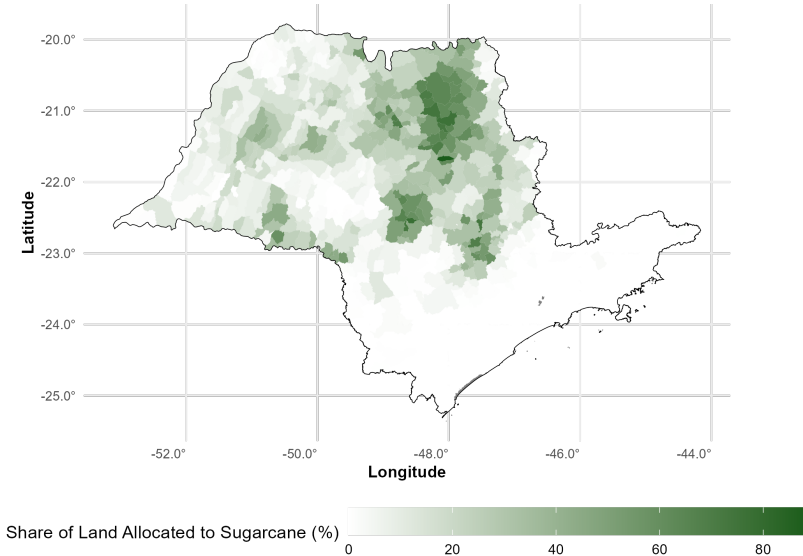
Area allowed to pre-harvest burning



Emissions from sugarcane agricultural waste burning



With a spatial heterogeneity in this reduction...



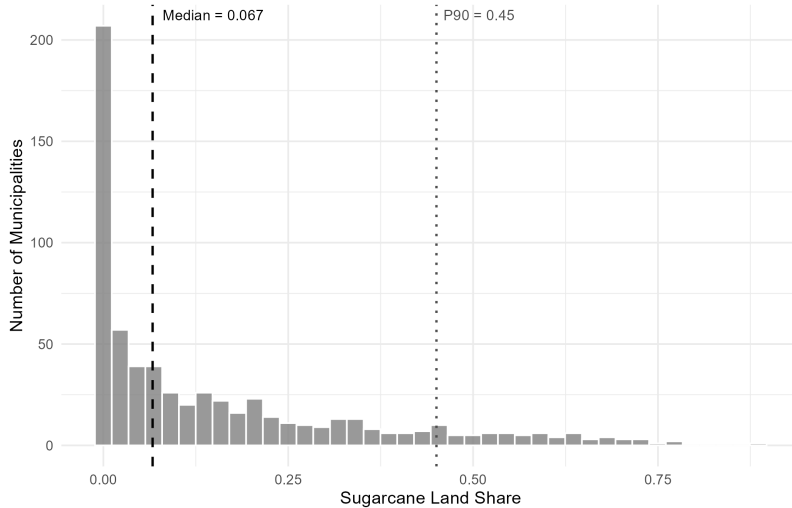
The Literature

- Rangel and Vogl [2019] – Review of Economics and Statistics: wind-based smoke exposure → lower birthweight, shorter gestation (São Paulo)
- Chagas et al. [2016] – Regional Science and Urban Economics: spatial DiD, sugarcane → respiratory hospitalizations, treatment is a binary median-split

Our contribution

- Broader outcomes: circulatory disease, infant mortality, a biologically-grounded congenital anomaly index
- Holding sugarcane production constant, what is the impact of reducing burning?
- Continuous rather than binary exposure, plus three sharper designs (season, wind, dose)

Heterogeneity matters



Data and Empirical Strategy

Data Sources

- **Hospitalizations:** SIH/SUS, 2008–2022 (ICD J00–J99, I00–I99)
- **Births:** SINASC, 2007–2023, $N \approx 9.7\text{M}$ individual records
- **Mortality:** SIM, 2002–2023
- **Treatment:** PAM-IBGE sugarcane land share, pre-2011 average
- **PM10/PM25:** CETESB monitoring network, 2003–2024
- **Fire:** MODIS MCD64A1 $\times$ MapBiomas LULC (cane pixels), Google Earth Engine
- **Wind:** NASA POWER/MERRA-2 reanalysis, Jul–Sep 2002–2022

All sources are administrative and cover the universe of events — no survey sampling error.

Combining the timing and the spatial heterogeneity

Difference-in-differences, continuous exposure

$$\ln(Y_{it}) = \alpha \cdot TC_{it} + \theta_i + \gamma_t + \epsilon_{it}$$

$$TC_{it} = Land_i \times D_{t>2010}, \quad Land_i = \text{pre-2011 sugarcane land share}$$

- θ_i, γ_t : municipality and year FE; regressions weighted by population (or births); SEs clustered by municipality
- $SD(Land_i) = 0.19$ — the unit for all “one-SD” effects below
- We do have specifications with individual births. So, the equation is analogous to the one described above but with maternal education, sex, marital status FE as well as

Results

Hospitalizations Fall for Respiratory & Circulatory Disease

$y = \ln(\text{hosp. per capita})$	Full (2008–2021)		Pre-Covid (2008–2019)	
	(1)	(2)	(3)	(4)
<i>Panel A: Respiratory (J00–J99)</i>				
Tech. Change	−0.380*** (0.114)	−0.369*** (0.112)	−0.287** (0.111)	−0.286** (0.111)
<i>Panel B: Circulatory (I00–I99)</i>				
Tech. Change	−0.238*** (0.078)	−0.224*** (0.075)	−0.234*** (0.078)	−0.222*** (0.075)
log(GDP/cap) control		X		X
Municipality & Year FE	X	X	X	X

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Weighted by population, clustered SEs.

- **1 SD more exposure** $\Rightarrow$ respiratory $\downarrow$ **7.2%**, circulatory $\downarrow$ **4.5%**; both stable when controlling for GDP/capita

Birth Outcomes: No Effect on Weight, Fewer Anomalies

	Any Anomaly (Q00–Q99)	Pol3 Index (NTD+Orofac.+Card.)	LBW (<2,500g)	VLBW (<1,500g)
Tech. Change	−0.006 66* (0.003 90)	−0.004 27** (0.001 82)	0.001 24 (0.002 00)	0.000 10 (0.000 84)
Baseline mean	11.45/1000	3.59/1000	9.26%	1.47%

Individual (Sinasc), $N = +9M$, Controls: Mun., year, educ., sex, marital, race...

- **No effect on birthweight** (LBW, VLBW)
- **Congenital anomalies fall 11%**; the **Pol3 pollution-linked index falls 22.6%** at 1 SD of exposure
- Pol3 = neural tube defects + orofacial clefts + cardiac malformations

Which Anomalies?

$y = \text{Pr}(\text{anomaly type})$	Neural Tube (Q00–Q07)	Orofacial (Q35–Q37)	Cardiac (Q20–Q26)	Pol3 Index
Tech. Change	−0.000 539** (0.000 221)	0.000 000 (0.000 156)	−0.003 846** (0.001 623)	−0.004 265** (0.001 823)
Baseline (per 1,000)	1.09	0.73	1.91	3.59
Relative effect (1 SD)	−9.4%	≈ 0	−38.3%	−22.6%

Mortality: A Dose-Response Gradient

Adults — no effect

	Respiratory	Circulatory
Exp	-0.028 (0.093)	-0.043 (0.071)

⇒ -0.5% and -0.8% at 1 SD —
statistically zero

Infants (<5) — significant

	Resp. mortality
Exp	-0.593* (0.344)

⇒ -11% at 1 SD; all-cause infant mortality:
null

Infants are physiologically more susceptible.

Is It Really the Smoke?

Less Fire and Less Particulate Matter

$$\ln(Y_{it}) = \alpha \cdot \underbrace{Land_i \times Post_t}_{\text{same treatment as before}} + \theta_i + \gamma_t + \varepsilon_{it}$$

$Y_{it} \in \{ \text{burned area on cane pixels, PM10, PM25} \}$

	Harvest (Jul-Sep)	Off-season (Jan-Mar)
<i>Panel A: Sugarcane burning (MODIS × MapBiomass)</i>		
$\ln(\text{burned area, cane pixels})$	-4.964*** (0.450)	—
<i>Panel B: Particulate matter (CETESB)</i>		
$\ln(\text{PM10})$	-0.094*** (0.010)	-0.005 (0.006)
$\ln(\text{PM25})$	-0.091*** (0.010)	-0.001 (0.006)

*** $p < 0.01$. Off-season burned area near-zero throughout (placebo).

- Harvest-season burned area on cane fields falls $\approx 66\%$ at mean exposure

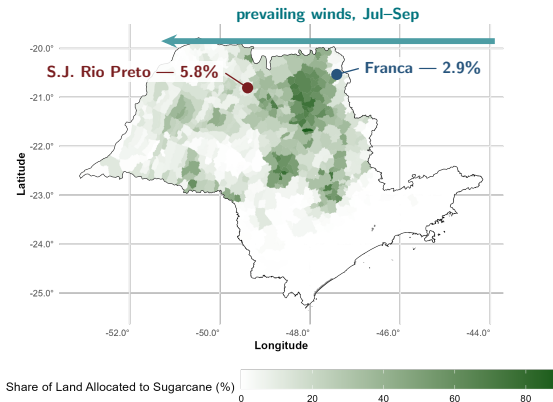
Health Gains Concentrate in Harvest Months

$$\ln Y_{itm} = \alpha \cdot Exp_{it} + \beta \cdot (Exp_{it} \times Harvest_m) + \theta_{i \times m} + \gamma_t + \varepsilon_{itm}$$

$y = \ln(\text{hosp.}/\text{pop})$	Respiratory	Circulatory
Exp (off-season)	-0.135*** (0.033)	-0.083** (0.028)
$Exp \times Harvest$ (Jul-Sep)	-0.069*** (0.011)	-0.014 (0.013)

Respiratory hospitalizations fall more when burning would have occurred. Circulatory shows no difference.

Why the wind matters



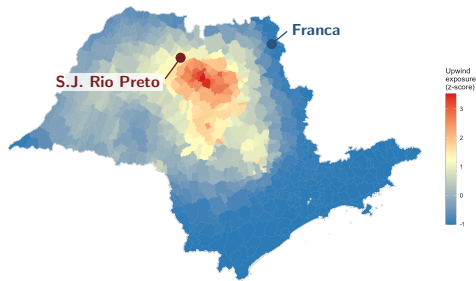
Franca and **São José do Rio Preto** grow almost no cane — 2.9% and 5.8% of their land.

Between them lies the dark core of the map: the Ribeirão Preto belt, where municipality after municipality puts half to three-quarters of its land under cane.

The wind runs east to west. Franca sits **upwind** — the smoke leaves it alone. Rio Preto sits **downwind** — it breathes a fire it never lit.

Robustness only — the main results do not depend on it. But the wind question is the fashion in environmental economics now, and we follow the fashion.

Smoke Travels with the Wind

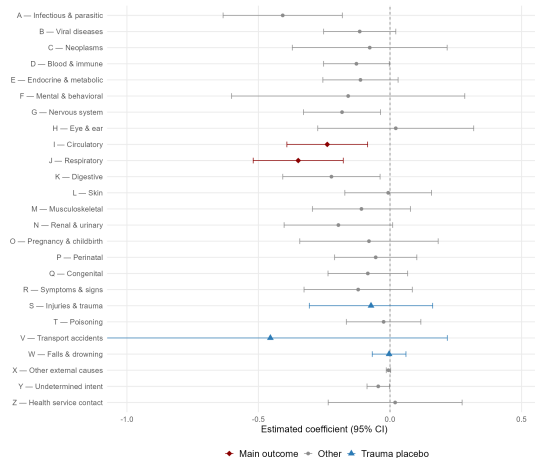


	Respiratory		Circulatory	
	Full	Pre-Cov.	Full	Pre-Cov.
Exp	−0.348***	−0.265***	−0.238***	−0.222***
WindExp	−0.035*	−0.027	−0.033	−0.011

Municipalities **downwind** of cane fields during the burning season gain more than upwind ones with identical own-exposure. Wind direction is plausibly exogenous to municipal health trends.

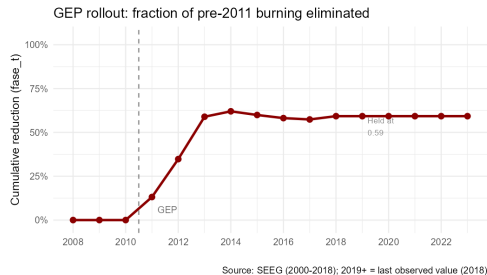
More Robustness

Placebo Across All ICD-10 Chapters



If the design were picking up general improvements in municipal health infrastructure, every chapter would move.

Continuous Treatment Intensity



	Respiratory	Circulatory
Binary (0/1)	−0.344***	−0.239***
Continuous ($\times fase_t$)	−0.618***	−0.393**

Replacing the post-2010 dummy with $fase_t$ — the fraction of burning emissions already eliminated — gives an implied effect at the mean of -0.346 , essentially the binary estimate.

Robustness: Excluding São Paulo City

	Respiratory		Circulatory	
	Full	Pre-Covid	Full	Pre-Covid
Baseline	-0.380***	-0.287**	-0.238***	-0.234***
Excl. SP city	-0.327**	-0.236*	-0.228**	-0.189**

São Paulo city is 12M people with zero cane share — a heavily weighted, mechanically untreated observation. Dropping it leaves the estimates essentially unchanged.

What Are the Stakes?

- Aggregating the hospitalization reduction over 2011–2021 across exposed municipalities: **≈118,000** fewer hospital admissions (71,000 respiratory + 47,000 circulatory)
- At 2019 SUS reimbursement rates: **R\$ 209 million** (USD 53M) in avoided hospitalization costs
- **≈1,640** fewer births with cardiac malformations (2011–2023) $\Rightarrow$ **R\$ 22 million** in avoided surgical costs

R\$ 232 million (USD 59 million) in avoided costs over a decade — a conservative *lower bound* that excludes outpatient care, productivity losses, lifetime disability transfers, and the value of infant lives saved.

Takeaway

- A gradual, phased ban on pre-harvest sugarcane burning — backed by regulation *and* voluntary export-market incentives — delivered large, well-identified health gains
- The pattern across outcomes traces a **dose-response gradient**: null adult mortality → significant infant mortality → significant hospitalizations → significant congenital anomalies, each margin defined by a different vulnerability threshold
- Relevant beyond São Paulo: Northeast Brazil (Pernambuco: >90% manual), India, Thailand, Florida still rely on pre-harvest burning

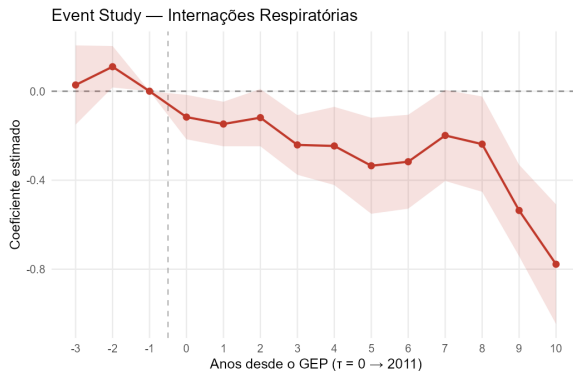
Thank you! Questions?

Backup Slides

Summary Statistics

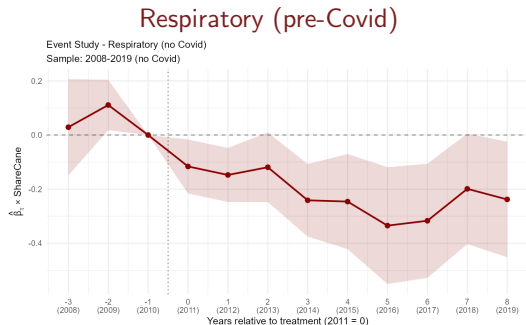
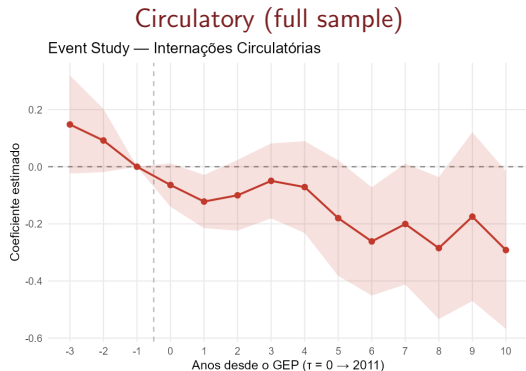
Variable	<i>N</i>	Mean	Median	SD	Max
Sugarcane land share (2002–2010)	642	0.150	0.067	0.190	0.886
Respiratory hosp. per 1,000 pop.	8,385	7.281	—	4.939	45.828
Circulatory hosp. per 1,000 pop.	8,385	6.470	—	3.085	25.760
Any anomaly (per 1,000 births)	11,882	18.662	—	91.807	1,000
Infant resp. mortality (per 1,000 births)	9,029	0.702	—	3.204	111.111

Identification: Parallel Trends Hold



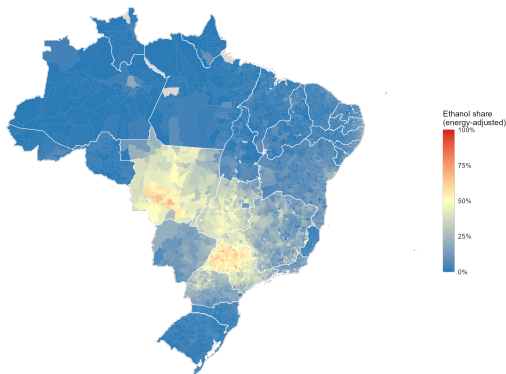
Pre-treatment coefficients ($\tau = -3, -2$) near zero and insignificant; post-treatment effects grow steadily as the GEP phases in — consistent with a gradual, accelerating rollout, not a one-time break.

Event Study — Circulatory and Pre-Covid Samples



The same pattern survives dropping the pandemic years, so the result is not a Covid artifact.

Backup: Ethanol Market Context



Energy-adjusted hydrated ethanol share in the fuel mix, 2020–2024. Hydrated ethanol competes directly with gasoline where transport costs from the cane belt are low.

Backup: Aggregate Birth Outcomes (Municipality-Year)

	$\ln(\text{birthweight})$	$\ln(\text{LBW})$	$\ln(\text{VLBW})$	$\ln(\text{anomalies})$
Exp	-0.002 (0.003)	0.027 (0.023)	-0.067 (0.056)	-0.795** (0.311)

Aggregate spec confirms the anomaly result; individual-level LPM (main text) additionally decomposes it by diagnostic group.

References I

- André LS Chagas, Carlos R Azzoni, and Alexandre N Almeida. A spatial difference-in-differences analysis of the impact of sugarcane production on respiratory diseases. *Regional Science and Urban Economics*, 59:24–36, 2016.
- C Austin Davis. Why did sugarcane growers suddenly adopt existing technology? Technical report, Working Paper - Yale Department of Economics, 2017.
- Marcos A Rangel and Tom S Vogl. Agricultural fires and health at birth. *Review of Economics and Statistics*, 101(4):616–630, 2019.